Do Aggregate Trade Restrictions Affect Sectors Equally? Evidence from High- and Low-GVC Industries Using a New Measure of Trade Restrictiveness

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1 Introduction

Global trade has become deeply interconnected through global value chains (GVCs), allowing countries to specialize across different stages of production. As Antràs (2020) defines it, a GVC is a series of production stages in which each stage adds value to the next. However, rising protectionism, trade tensions, and industrial policies have increasingly challenged this openness in recent years, driven by geoeconomic rivalries, policy uncertainty, and related factors (see Jiang et al. 2023; Gourinchas 2022; Fajgelbaum and Khandelwal 2022; Campos, Pienknagura, and Timini 2023). While aggregate trade restrictions generally depress trade, their effects likely vary across sectors: industries that are highly integrated into GVCs can adapt through flexible sourcing, whereas those trading primarily in final goods remain more vulnerable to policy barriers.

Traditional measures of trade restrictiveness, such as tariffs or dummy variables for trade agreements, often fail to capture the full range of policy instruments that influence trade flows. Many modern restrictions operate through non-tariff channels such as regulatory barriers, subsidies, or export controls. This paper exploits the recently developed Measure of Aggregate Trade Restrictions (MATR) by Estefania-Flores et al. (2024) from the IMF, which offers a comprehensive, time-varying index that aggregates both tariff and non-tariff barriers. Despite its growing use in macro-trade analysis, most studies have focused on aggregate trade, leaving a gap in the literature of how these restrictions affect specific industries with varying GVC intensity.

This paper addresses this gap by asking a simple but important question: *Do aggregate trade restrictions affect all sectors equally?* Using MATR as a unified policy indicator, we combine data

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from the CEPII gravity database replicated from Campos et al. (2023) with UN Comtrade bilateral exports at the HS-2 level from 1989 to 2019. The empirical analysis estimates a structural gravity model with high-dimensional fixed effects to identify how policy restrictiveness affects trade across both high-GVC and low-GVC industries. High-GVC sectors (such as electronics and machinery) involve complex cross-border production networks, while low-GVC sectors (such as fuels and agricultural goods) rely more on direct, final-good trade.

The paper makes three key contributions. First, it provides the first cross-sectoral application of the MATR index in a consistent gravity-model framework. Second, it distinguishes the heterogeneous effects of trade restrictiveness across industries with different GVC intensities, linking policy shocks to production structure. Third, it contributes to the debate on whether recent trade restrictions, such as those between the United States and China, disproportionately harm industries that are less globally integrated.

The results suggest that the effects of trade restrictions are not uniform across sectors. High-GVC industries, such as electronics, appear relatively less sensitive to MATR, consistent with the idea that intermediate goods trade is often exempted or rerouted within production networks. In contrast, low-GVC sectors such as fuels show stronger negative responses, indicating that aggregate restrictions tend to hit final-goods trade more directly. Overall, trade restrictions still matter, but their strength depends on how deeply sectors are in the global production networks.

2 Literature Review

Early empirical research primarily emphasized tariffs as the dominant source of trade costs. Looi Kee, Nicita, and Olarreaga (2009) provide one of the first systematic quantifications of trade restrictiveness across countries, showing that non-tariff barriers (NTBs) often outweigh tariff effects. Since then, the composition of trade restrictions has evolved, with policy attention shifting toward regulatory, technical, and administrative barriers. The introduction of the Measure of Aggregate Trade Restrictions (MATR) by Estefania-Flores et al. (2024) provides a more comprehensive view, aggregating tariffs, NTBs, and quantitative restrictions into a single index. A recent application of this measure by Campos et al. (2023) using a gravity framework finds that higher policy restrictions reduce bilateral trade by about 6% on average.

At the same time, the rise of GVCs have fundamentally changed how countries engage in trade. As Johnson (2018) show, an increasing share of value added now crosses borders multiple times, meaning that production is far more interconnected and less sensitive to traditional trade costs like tariffs. Deeper GVC participation also shapes governments' trade policies (i.e., countries that rely more on imported inputs tend to maintain lower tariffs and avoid temporary protection measures (Blanchard, Bown, and Johnson 2016)). At the firm level, GVCs offer flexibility. Gereffi, Lim, and Lee (2021) find that companies often respond to trade restrictions by reorganizing their production networks, moving suppliers, or upgrading to higher-value activities. This adaptability helps highly integrated sectors absorb policy shocks more easily, while sectors trading mostly in final goods remain more vulnerable. These differences motivate our comparison between high-GVC sectors (e.g., `HS85`, `HS84`) and low-GVC sectors (e.g., `HS02`, `HS27`) to understand how trade restrictions affect them differently.

Despite these advances, few studies have brought these two strands of research together. Most

existing work either looks at aggregate trade responses to policy shocks (e.g., Campos et al. 2023) or examines GVC participation without considering how broad policy restrictions shape trade patterns (e.g., Johnson 2018). This paper bridges the two by estimating a structural gravity model with MATR as the central explanatory variable, comparing how trade restrictiveness affects high- and low-GVC sectors across countries and over time. Specifically, we address three key questions: (1) How does MATR affect aggregate trade and sectoral trade? (2) Are the effects stronger for sectors trading final goods (e.g., agriculture) than intermediate goods (e.g., electronics)? (3) How have these relationships evolved over time?

3 Data and Methodology

3.1 Data

The data builds upon the replication dataset of Campos et al. (2023), merged with sector-level trade data from UN Comtrade to explore heterogeneous effects of trade restrictions across sectors with varying degrees of global value chain (GVC) participation. Sectors are classified according to their GVC integration. High-GVC sectors—`HS85` (Electrical machinery and electronics) and `HS84` (Machinery and mechanical appliances)—feature multi-stage cross-border production and high intermediate-input intensity. In contrast, low-GVC sectors—`HS02` (Meat and edible offal), `HS10` (Cereals), and `HS27` (Mineral fuels and oils)—primarily trade final goods or raw commodities. `HS90` (Optical and precision instruments) serves as a medium-GVC benchmark, given its mix of intermediate and final trade characteristics.[1]

The final dataset spans from 1989–2019, covering over 150 countries and includes trade flows by sector from the UN Comtrade Harmonized System (HS) classification. Table 1 shows the descriptive statistics. The variable of interest, Measure of Aggregate Trade Restrictions (`MATR`) described in Table 10, shows wide variation across country pairs and years. Sectoral trade values (e.g., `HS85`, `HS84`, and `HS27`) also vary substantially, reflecting differences in GVC participation. Policy variables such as `GATT/WTO` and `RTA` dummies show balanced coverage, while the relatively low mean of the `EU` dummy highlights the limited share of intra-EU trade in the global sample.

Summary plots of global trade patterns and `MATR` over time are presented in Appendix 6. Figure 1 and Figure 2 illustrate the evolution of global and sectoral trade, where electronics and machinery remain dominant. Figure 3 displays sectoral trade by economic blocs, suggesting a possible disruption around the 2008 Global Financial Crisis (GFC). Finally, Figure 4 shows a gradual decline in `MATR` values over time, reflecting a general trend toward more open global trade.

While the merged dataset provides broad coverage, several limitations should be noted. First, not all countries report detailed sectoral exports every year, resulting in an unbalanced panel. Second, discrepancies between reported and mirror trade values occasionally occur in UN Comtrade data. Third, `MATR` captures trade restrictiveness at the national level and may not fully reflect sector-specific policies. Despite these caveats, the dataset's wide cross-country and time variation in `MATR` offers sufficient scope for identifying its effects on trade flows.

1. Sector classifications are approximated using the OECD–WTO Trade in Value Added (TiVA) framework, which reports the share of foreign value added embodied in exports. Sectors with higher foreign value-added shares are identified as high-GVC, while those with lower shares are classified as low-GVC.

Table 1: Descriptive Statistics.

height**Variable**	**Source**	**N**	**Mean**	**Std. dev.**	**Min**	**Max**
Measure of Aggregate Trade Restrictions (`matr`)	Estefania-Flores et al. (2024)	717 963	10.42	4.80	0	21.00
MATR (non-tariff)	Estefania-Flores et al. (2024)	717 963	9.42	4.88	0	20.00
Aggregate Bilateral Trade (`trade`)	CEPII gravity database	470 703	3,159,300.59	118,079,355.71	0	1.99E+10
`HS85` (Electrical machinery and electronics)	UN Comtrade (HS 2-digit)	303 050	126,497.58	1,763,241.32	0	1.98E+08
`HS84` (Machinery, mechanical appliances)	UN Comtrade (HS 2-digit)	303 050	123,553.68	1,166,771.63	0	1.03E+08
`HS27` (Mineral fuels, oils)	UN Comtrade (HS 2-digit)	303 050	89,706.02	1,060,883.15	0	1.21E+08
`HS90` (Optical and precision instruments)	UN Comtrade (HS 2-digit)	303 050	29,925.58	313,858.48	0	2.38E+07
`HS02` (Meat and edible offal)	UN Comtrade (HS 2-digit)	303 050	6,736.13	69,965.68	0	4.54E+06
`HS10` (Cereals)	UN Comtrade (HS 2-digit)	303 050	5,651.83	60,187.21	0	5.89E+06
`GATT` dummy	CEPII gravity database	717 963	0.48	0.50	0	1
`WTO` dummy	CEPII gravity database	717 963	0.55	0.50	0	1
`GATT/WTO` combined dummy	CEPII gravity database	717 963	0.63	0.48	0	1
`RTA` dummy	CEPII gravity database	716 429	0.12	0.32	0	1
`EU` dummy	CEPII gravity database	717 963	0.02	0.14	0	1

Notes: Trade values are in nominal (thousands of USD). CEPII variables follow Campos et al. (2023). HS variables are constructed from UN Comtrade data at the HS 2-digit level. Sample sizes (`N`) differ across variables due to missing values in the merged dataset. For estimation (PPML regressions), missing (`NA`) values are replaced with zeros to retain zero-trade observations.

3.2 Empirical Specification

The model follows the benchmark gravity framework of Larch, Shikher, and Yotov (2025) and extending the works of Campos et al. (2023), where we estimate the effect of aggregate trade restrictions on bilateral trade flows across sectors. The recommended model is defined as:

$$X_{ij,t}^{s} = \exp\left[\pi_{i,t} + \chi_{j,t} + \gamma_{ij} + \beta_1 \texttt{MATR}_{ijt} + \mathbf{Z}'_{ijt}\psi\right] \times \epsilon_{ij,t}, \tag{1}$$

where $X_{ij,t}^{s}$ denotes bilateral exports from country i to country j in sector s at time t, measured in nominal USD. The superscript s identifies the sector (for example, `HS85` for electronics or `HS02` for meat products), allowing heterogeneous policy effects to be estimated across industries. The key explanatory variable $\texttt{MATR}_{ijt}$ is the IMF's Measure of Aggregate Trade Restrictions, where one unit change in `MATR` affects trade by approximately $100 \times [e^{\hat{\beta}_1} - 1]$ percent.

$\mathbf{Z}'_{ijt}$ is a vector of time-varying controls that include other trade-policy indicators such as GATT/WTO membership, participation in regional trade agreements (RTA), and European Union (EU) integration. The terms $\pi_{i,t}$ and $\chi_{j,t}$ denote exporter-time and importer-time fixed effects, respectively, which absorb country-specific factors such as GDP, productivity, and macroeconomic shocks that vary over time. The pair-specific effect γ_{ij} captures all time-invariant bilateral factors such as distance, common language, colonial ties, and geography.

Unlike the full benchmark model, our preferred specification adopts a hybrid fixed-effect design using directional pair γ_{ij} and year δ_t fixed effects only. When including exporter-year and importer-year fixed effects, given our dataset, the estimates would fully absorb `MATR` in the sector-level setting, preventing identification of its coefficient. This hybrid approach, therefore, provides a practical workaround that preserves within-pair and time variation while controlling for common global shocks.

Although this comes at the cost of losing some structural properties, such as the full control for unobservable multilateral resistance terms (see Anderson and Van Wincoop (2003)Larch, Shikher, and Yotov (2025)), it remains the most suitable specification for our disaggregated dataset. Rather than explaining overall trade levels, the model asks: *within a given country pair, how do changes in trade restrictiveness affect sectoral trade over time?*. The baseline model to be estimated is the following:

$$X_{ij,t}^{s} = \exp\left[\delta_t + \gamma_{ij} + \beta_1 \texttt{MATR}_{ijt} + \mathbf{Z}'_{ijt}\psi\right] \times \epsilon_{ij,t}, \tag{2}$$

The model is estimated using the Pseudo-Poisson Maximum Likelihood (PPML) estimator to handle zero trade flows and heteroskedasticity. All regressions are estimated using the `ppmlhdfe` command in `Stata`, which efficiently absorbs multiple high-dimensional fixed effects and clusters standard errors by exporter, importer, and year.

We begin by estimating the baseline model without any additional controls. Next, as part of a series of robustness checks, we add policy controls, test a version of `MATR` that excludes tariff components, and focus on the U.S.–China trade relationship to see how results hold in a key bilateral context. Finally, we explore how the effects of trade restrictions have changed over time, asking whether trade has become less sensitive to policy barriers across different sectors.

4 Results

4.1 Baseline Results

Table 2 reports the baseline results showing that higher trade restrictiveness significantly reduces bilateral trade flows across all sectors which is consistent with the theory. A one-unit increase in `MATR` reduces trade by approximately $100 \times (e^{\hat{\beta}} - 1)$ percent (e.g., around -7.7% for total trade). The negative and relatively similar magnitudes across sectors suggest that aggregate trade restrictions have a broadly consistent dampening effect, regardless whether they are high or low in GVC. However, these results may be confounded by other trade agreements or institutional factors which

masks the effect of trade restrictions which will be addressed in the following subsection.

Table 2: `ppmlhdfe` Gravity Model Estimates – Baseline Results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	Medium	Low	Low	Low
`MATR`	−0.080***	−0.115**	−0.101***	−0.123***	−0.097***	−0.065*	−0.109***
	(0.013)	(0.051)	(0.034)	(0.044)	(0.031)	(0.037)	(0.031)
Constant	20.571***	16.150***	15.577***	14.557***	15.788***	12.507***	13.179***
	(0.024)	(0.402)	(0.239)	(0.282)	(0.279)	(0.320)	(0.207)
Observations	6.70e+05	4.81e+05	4.90e+05	4.27e+05	3.53e+05	2.57e+05	2.29e+05
Pseudo R^2	0.990	0.965	0.970	0.973	0.936	0.898	0.940
γ_{ij} (Directional Pair FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
δ_t (Year FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The specification includes directional country-pair and year fixed effects, controlling for time-invariant bilateral trade costs and global time shocks. Standard errors (in parentheses) are clustered by exporter, importer, and year.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Robustness Checks Results

Table 3 shows that even after adding policy controls, `MATR` remains negative, significant, but lower in magnitude for most sectors, confirming theoretical predictions from structural gravity and prior findings [2]. However, the effects on electronics (`HS85`) and cereals (`HS10`) becomes insignificant, which could plausibly reflect that electronics—being highly integrated in global value chains—are more insulated from policy shocks, while cereals, traded mainly as final goods, may respond differently due to agricultural and food security policies. Policy controls effects are as expected, positive across

2. see e.g., Anderson and Van Wincoop 2003;Campos et al. 2023

sectors, with some are statistically insignificant, notably, mineral fuels and oils (HS27).

Table 3: ppmlhdfe Gravity Model Estimates – With Policy Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	Medium	Low	Low	Low
MATR	−0.048***	−0.073	−0.069**	−0.096**	−0.096***	−0.051	−0.096***
	(0.014)	(0.050)	(0.032)	(0.047)	(0.030)	(0.036)	(0.032)
TA	0.268***	0.179*	0.172**	0.266***	0.066	0.173	0.317*
	(0.056)	(0.107)	(0.078)	(0.045)	(0.102)	(0.161)	(0.166)
GATT/WTO	0.770***	1.545***	1.158***	1.125***	0.044	0.513**	0.137
	(0.156)	(0.297)	(0.356)	(0.261)	(0.100)	(0.227)	(0.369)
EU	0.511***	0.951***	0.865***	0.742***	−0.111	1.479***	0.743
	(0.156)	(0.208)	(0.180)	(0.196)	(0.155)	(0.299)	(0.497)
Constant	20.324***	14.089***	13.955***	13.020***	15.721***	11.616***	12.483***
	(0.033)	(0.553)	(0.447)	(0.405)	(0.280)	(0.419)	(0.471)
Observations	6.69e+05	4.80e+05	4.89e+05	4.26e+05	3.52e+05	2.57e+05	2.29e+05
Pseudo R^2	0.990	0.972	0.975	0.976	0.936	0.901	0.941
γ_{ij} (Directional Pair FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
δ_t (Year FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: All specifications include directional country-pair and year fixed effects, controlling for time-invariant bilateral trade costs and global time shocks. Standard errors (in parentheses) are clustered by exporter, importer, and year.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Nevertheless, we carried out an additional robustness check by excluding tariff-measures from our variable of interest, as shown in Table 6. The results remain broadly consistent: higher non-tariff trade restrictions still reduce bilateral trade. Interestingly, once tariffs are removed, the estimated effect becomes statistically significant at the 10% level for electronics (HS85), which suggests that non-tariff measures — such as technical barriers to trade (TBTs), sanitary & phytosanitary standards (SPS), or product-specific compliance requirements — may matter especially for sectors embedded in global value chains where intermediate inputs cross borders multiple times. Meanwhile, the low-GVC sectors (e.g., cereals) still show no significant effect.

When we restrict the sample to trade involving the U.S. and China as both as importers and exporters, the results in Table 7 show distinct patterns across sectors. High-GVC industries such as electronics and machinery display insignificant effects, suggesting that global production networks help cushion them from trade restrictions. In contrast, low-GVC sectors like mineral fuels remain highly sensitive, reflecting their limited ability to adjust supply chains when policy barriers rise.

4.3 Globalization and the Evolution of Trade Sensitivity

As mentioned in the data sub-section, the trend in the Measure of Aggregate Trade Restrictions (MATR) suggests that the sensitivity of trade to policy barriers has shifted over time. Table 9 presents estimates capturing how trade responsiveness to MATR evolved with globalization. The baseline coefficients remain negative and significant across sectors, confirming that restrictions reduce trade. Furthermore, the positive interaction terms indicate that this negative effect has weakened over

time, except for mineral fuels, a sector that remains sensitive to trade restrictions across the years.

This pattern aligns with the idea that globalization has made trade increasingly resilient to policy frictions. Table 9 summarizes how the effect of `MATR` evolved before and after 2009. The post-GFC effects are calculated as $\beta_1 + \beta_2$, where β_1 represents the pre-GFC coefficient and β_2 captures the additional change after 2009.[3] It is important to note that the Global Financial Crisis (GFC) serves here as a convenient time marker rather than a strict structural break. The estimates show a gradual weakening in the negative impact of trade restrictions across most sectors. High-GVC sectors such as `HS85` (electronics) and `HS84` (machinery) appear less sensitive to policy barriers, reflecting their deeper integration in global production networks. In contrast, low-GVC sectors like `HS27` (fuels) and `HS02` (meat) remain more exposed to restrictive measures, while `HS10` (cereals) shows the largest improvement, possibly reflecting evolving supply chains and greater participation in global trade.

Table 4: `ppmlhdfe` Gravity Model Estimates – Globalization and Time Trends (with Controls)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	High	Low	Low	Low
`MATR` (β_1)	−0.062***	−0.148***	−0.149***	−0.175***	−0.109***	−0.147***	−0.135***
	(0.017)	(0.052)	(0.036)	(0.041)	(0.031)	(0.035)	(0.021)
`Post2009×MATR` (β_2)	−0.009	0.021	0.012	0.025**	−0.004	−0.006	−0.007
	(0.010)	(0.015)	(0.014)	(0.010)	(0.015)	(0.015)	(0.011)
`MATR×Trend` (β_3)	0.002***	0.003***	0.005***	0.004***	0.001	0.006***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)
`TA`	0.139***	0.144	0.121*	0.196***	0.060	0.237	0.318**
	(0.046)	(0.117)	(0.063)	(0.030)	(0.112)	(0.172)	(0.139)
`GATT/WTO`	0.563***	1.306***	0.891***	0.902***	0.016	0.201	−0.005
	(0.190)	(0.270)	(0.220)	(0.259)	(0.105)	(0.158)	(0.357)
`EU`	0.413***	0.878***	0.770***	0.613***	−0.086	1.453***	0.710
	(0.152)	(0.220)	(0.170)	(0.172)	(0.161)	(0.260)	(0.477)
Constant	20.348***	14.275***	14.096***	13.199***	15.670***	11.624***	12.346***
	(0.039)	(0.368)	(0.256)	(0.338)	(0.326)	(0.236)	(0.392)
Observations	6.69e+05	4.80e+05	4.89e+05	4.26e+05	3.52e+05	2.57e+05	2.29e+05
Pseudo R^2	0.990	0.975	0.978	0.978	0.936	0.906	0.943
γ_{ij} (Directional Pair FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
δ_t (Year FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: All specifications include directional country-pair and year fixed effects, controlling for time-invariant bilateral trade costs and global time shocks. The coefficients represent the effects of `MATR` on trade, adjusted for major policy variables (`TA`, `GATT/WTO`, and `EU`). Positive and significant time-trend interactions (β_3) indicate that the negative impact of trade restrictiveness has gradually weakened over time, consistent with global trade integration. Standard errors (in parentheses) are clustered by exporter, importer, and year.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3. The marginal post-GFC effect of `MATR` is given by $\partial X_{ijt}/\partial \texttt{MATR} = \beta_1 + \beta_2$.

5 Discussion

Table 5 summarizes the main findings. Consistent with gravity theory, higher trade restrictiveness (MATR) significantly reduces bilateral trade across all sectors. However, once policy controls are added, the effect becomes insignificant for electronics (HS85) and cereals (HS10), suggesting that globally integrated sectors are more insulated from policy shocks, while agriculture may be shaped by distinct food security and subsidy policies.

Table 5: Summary of Estimated Effects of Trade Restrictiveness (MATR) Across Scenarios

	Baseline	With Controls	MATR_NT	U.S.–China	Pre-GFC	Post-GFC
Trade (Aggregate)	−0.080***	−0.048***	−0.050***	−0.039*	−0.049***	−0.033***
HS85 (High-GVC)	−0.115**	−0.073	−0.086*	−0.029	−0.105***	−0.042**
HS84 (High-GVC)	−0.101***	−0.069**	−0.085***	−0.052	−0.096***	−0.025**
HS90 (Medium)	−0.123***	−0.096**	−0.104**	−0.056	−0.128***	−0.058***
HS27 (Low-GVC)	−0.097***	−0.096***	−0.092***	−0.165***	−0.098***	−0.090***
HS10 (Low-GVC)	−0.065*	−0.051	−0.037	−0.005	−0.082***	−0.011
HS02 (Low-GVC)	−0.109***	−0.096***	−0.097***	+0.013	−0.095***	−0.046***

When MATR is re-estimated excluding tariffs (MATRNT), electronics become significantly affected, indicating that non-tariff measures—such as standards and compliance requirements—pose meaningful barriers for intermediate trade within global value chains.

In the U.S.–China sample, only mineral fuels (HS27) remain sensitive to trade restrictiveness, reflecting how energy trade is driven more by strategic and geopolitical factors than by production-network dynamics.

Finally, over time, the negative effect of MATR has weakened, particularly after 2009, consistent with deeper GVC integration and improved supply-chain adaptability. High-GVC sectors show growing resilience, while low-GVC sectors, especially fuels, remain vulnerable. Overall, aggregate trade restrictions still matter, but their strength now depends on how deeply sectors are embedded in global production networks.

6 Conclusion

This paper examined how aggregate trade restrictiveness, measured by the Measure of Aggregate Trade Restrictions (MATR), influences global trade across sectors with differing levels of global value chain (GVC) integration. Motivated by the rise of protectionism and policy uncertainty, we asked whether GVC participation moderates the sensitivity of trade to restrictive policies.

Using a structural gravity model estimated with ppmlhdfe, we found that trade restrictiveness significantly reduces bilateral trade, though the effects vary by sector. High-GVC industries such as electronics and machinery are less sensitive to policy shocks, reflecting their flexibility in sourcing and production, while low-GVC sectors—especially fuels and agricultural goods—remain more exposed. Over time, the impact of restrictions has weakened, suggesting that globalization has increased the resilience of trade to policy frictions.

Although the MATR index provides a comprehensive and comparable measure, it remains an

aggregate indicator that may mask product- or partner-specific heterogeneity. Our analysis is also correlational rather than causal, as possible endogeneity between trade and policy behavior is not explicitly addressed. Additionally, the model specification could be refined further. Despite these caveats, the study extends previous work of Campos et al. (2023) on aggregate trade restrictiveness to a sectoral context and provides new insights into how GVC integration shapes the effects of trade policy.

Overall, the findings suggest that traditional protectionist policies are becoming less effective in an era of fragmented global production. Rather than relying on broad restrictions, policymakers should adopt more targeted and cooperative approaches that strengthen supply-chain resilience and preserve open trade frameworks such as the WTO and regional agreements. Future studies could build on this work by using causal methods and more detailed, sector-level measures of trade restrictiveness to better understand how policy shapes global production networks.

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Appendix

Robustness Checks

Table 6: `ppmlhdfe` Gravity Model Estimates – Using Non-Tariff Trade Restrictions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	Medium	Low	Low	Low
`MATR_NT`	−0.050***	−0.086*	−0.085***	−0.104**	−0.092***	−0.037	−0.097***
	(0.014)	(0.044)	(0.028)	(0.044)	(0.031)	(0.031)	(0.034)
`TA`	0.262***	0.176*	0.168**	0.262***	0.077	0.175	0.317*
	(0.052)	(0.106)	(0.076)	(0.042)	(0.106)	(0.161)	(0.164)
`GATT/WTO`	0.764***	1.538***	1.150***	1.118***	0.059	0.521**	0.143
	(0.157)	(0.293)	(0.350)	(0.259)	(0.101)	(0.225)	(0.368)
`EU`	0.503***	0.924***	0.826***	0.726***	−0.089	1.530***	0.754
	(0.157)	(0.210)	(0.176)	(0.194)	(0.158)	(0.293)	(0.498)
Constant	20.320***	14.120***	14.009***	12.990***	15.550***	11.432***	12.388***
	(0.036)	(0.452)	(0.381)	(0.353)	(0.244)	(0.343)	(0.454)
Observations	6.69e+05	4.80e+05	4.89e+05	4.26e+05	3.52e+05	2.57e+05	2.29e+05
Pseudo R^2	0.990	0.972	0.975	0.976	0.936	0.901	0.941
γ_{ij} (Directional Pair FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
δ_t (Year FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: All specifications include directional country-pair and year fixed effects, controlling for time-invariant bilateral trade costs and global time shocks. `MATR_NT` captures only non-tariff trade restrictions. Sectors are grouped by their approximate GVC intensity based on OECD–WTO TiVA classifications. Standard errors (in parentheses) are clustered by exporter, importer, and year.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: `ppmlhdfe` Gravity Model Estimates – U.S.–China Sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	Medium	Low	Low	Low
`MATR`	−0.039*	−0.029	−0.052	−0.056	−0.165***	−0.005	0.013
	(0.022)	(0.051)	(0.054)	(0.058)	(0.039)	(0.024)	(0.044)
`TA`	0.149	0.310***	0.135	0.402***	0.228	0.340	0.398
	(0.114)	(0.111)	(0.153)	(0.064)	(0.147)	(0.217)	(0.282)
`GATT/WTO`	0.986***	1.992***	1.735***	1.528***	0.265	−0.006	0.235
	(0.147)	(0.197)	(0.344)	(0.115)	(0.273)	(0.245)	(0.497)
Constant	21.772***	14.744***	14.727***	13.584***	16.832***	12.991***	12.861***
	(0.057)	(0.562)	(0.595)	(0.482)	(0.427)	(0.134)	(0.527)
Observations	18,688	18,147	18,079	17,814	17,288	15,973	15,055
Pseudo R^2	0.989	0.975	0.970	0.970	0.925	0.922	0.928
γ_{ij} (Directional Pair FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
δ_t (Year FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Sample restricted to trade flows involving the U.S. and China as either exporters or importers. All specifications include directional country-pair and year fixed effects, controlling for time-invariant bilateral trade costs and common time shocks. Sectors are grouped by their approximate GVC intensity based on OECD–WTO TiVA classifications. Standard errors (in parentheses) are clustered by exporter, importer, and year.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: `ppmlhdfe` Gravity Model Estimates – Pre- and Post-GFC Interaction Effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	High	Low	Low	Low
`MATR`	−0.049***	−0.105***	−0.096***	−0.128***	−0.098***	−0.082***	−0.095***
	(0.014)	(0.039)	(0.028)	(0.041)	(0.028)	(0.027)	(0.020)
`post2009=1 # MATR`	0.016***	0.063**	0.071***	0.070***	0.008	0.071***	0.049***
	(0.004)	(0.027)	(0.018)	(0.011)	(0.021)	(0.020)	(0.009)
`TA`	0.198***	0.146	0.129*	0.208***	0.061	0.224	0.307**
	(0.059)	(0.110)	(0.066)	(0.031)	(0.110)	(0.167)	(0.141)
`GATT/WTO`	0.667***	1.375***	0.980***	0.963***	0.026	0.264	0.032
	(0.178)	(0.252)	(0.253)	(0.245)	(0.100)	(0.163)	(0.355)
`EU`	0.459***	0.910***	0.811***	0.654***	−0.099	1.469***	0.736
	(0.151)	(0.210)	(0.167)	(0.175)	(0.158)	(0.262)	(0.481)
Constant	20.341***	14.224***	14.061***	13.149***	15.705***	11.692***	12.389***
	(0.036)	(0.384)	(0.290)	(0.338)	(0.303)	(0.276)	(0.404)
Observations	6.69e+05	4.80e+05	4.89e+05	4.26e+05	3.52e+05	2.57e+05	2.29e+05
Pseudo R^2	0.990	0.974	0.977	0.978	0.936	0.904	0.942
γ_{ij} (Directional Pair FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
δ_t (Year FE)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Interaction terms allow the effect of trade restrictiveness (`MATR`) to differ before and after the 2008 Global Financial Crisis (`post2009`). All specifications include directional country-pair and year fixed effects, controlling for time-invariant bilateral trade costs and global time shocks. Sectors are grouped by their approximate GVC intensity based on OECD–WTO TiVA classifications. Standard errors (in parentheses) are clustered by exporter, importer, and year.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Estimated Difference in `MATR` Effects Before and After the GFC

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable:	**Trade**	**HS85**	**HS84**	**HS90**	**HS27**	**HS10**	**HS02**
GVC Intensity:	–	High	High	High	Low	Low	Low
Pre-GFC (β_1)	−0.049***	−0.105***	−0.096***	−0.128***	−0.098***	−0.082***	−0.095***
Post-GFC ($\beta_1 + \beta_2$)	−0.033***	−0.042**	−0.025**	−0.058***	−0.090***	−0.011	−0.046***
Change (β_2)	+0.016***	+0.063**	+0.071***	+0.070***	+0.008	+0.071***	+0.049***
Observations	6.69e+05	4.80e+05	4.89e+05	4.26e+05	3.52e+05	2.57e+05	2.29e+05

Notes: Values are computed based on coefficients from Table 8. Pre-GFC effects correspond to β_1, while post-GFC effects are the sum $\beta_1 + \beta_2$. A smaller (less negative) post-GFC coefficient implies that trade became less sensitive to policy restrictiveness after 2009.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figures

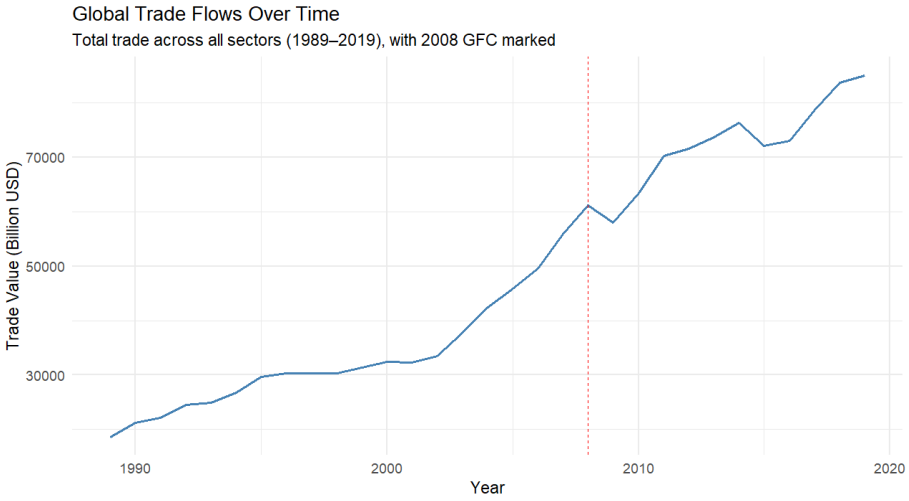


Figure 1: Global Trade Flows (1989−2019)

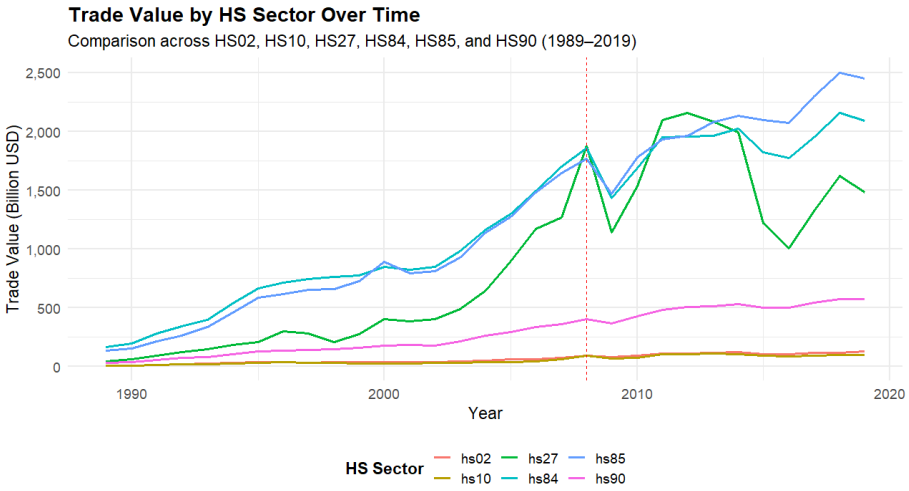


Figure 2: Global Trade Flows by Selected Sectors (1989−2019)

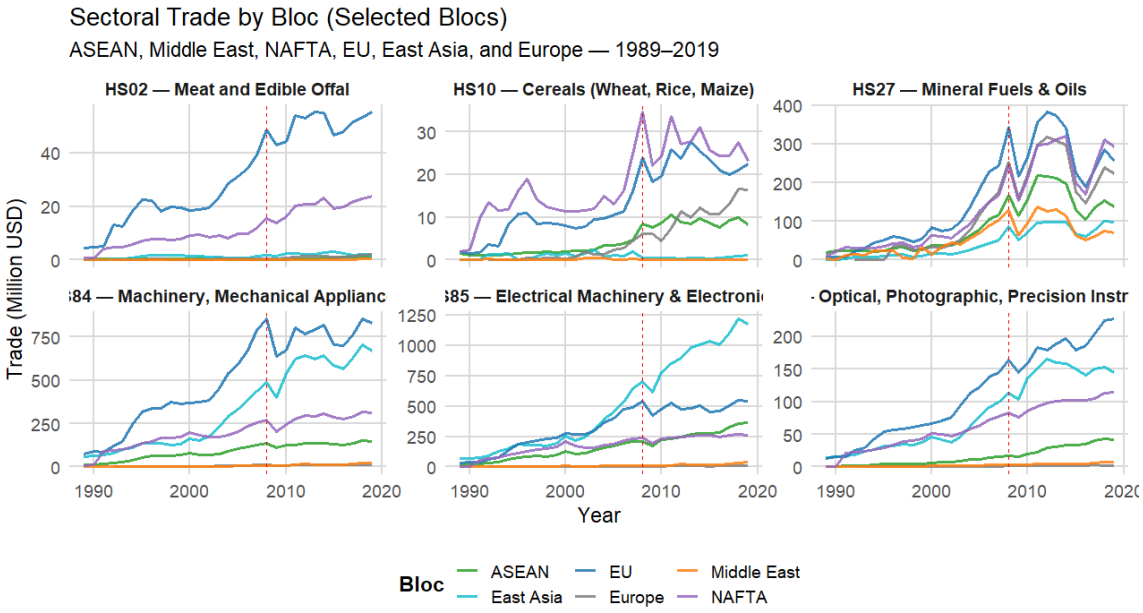


Figure 3: Trade Flows on Selected Sectors by Blocs (1989−2019)

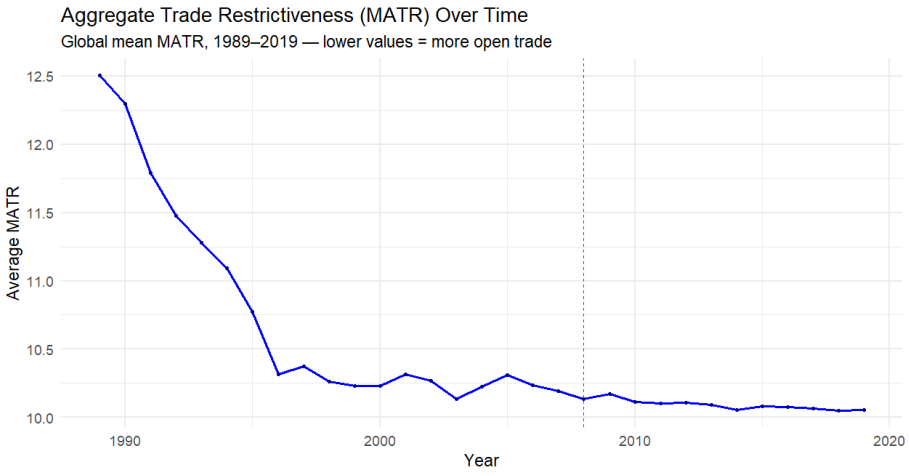


Figure 4: Average Measure of Aggregate Trade Over Time (1989−2019)

Data Description

Table 10: MATR Components

MATR Component	Subcomponents
II. Exchange measures	II.A. Restrictions and/or multiple currency practices
	II.B. Exchange measures imposed for security reasons
IV. Restrictions to payments	IV.A. Prescription of currency requirements
	IV.B. Payments arrangements
	IV.C. Administration of control
	IV.D. Payment arrears
	IV.F. Controls on exports and imports of banknotes
VII. Import Restrictions	VII.A. Foreign exchange budget
	VII.B. Financing requirements for imports
	VII.C. Documentation requirements for release of forex for imports
	VII.D. Import licenses and other nontariff measures
	VII.E. Import taxes and/or tariffs
	VII.F. State Import Monopoly
VIII. Export Restrictions	VIII.A. Repatriation requirements
	VIII.B. Financing requirements
	VIII.C. Documentation requirements
	VIII.D. Export licenses
	VIII.E. Export taxes
IX. Payments	IX.A. Payments for Invisibles, Transfers & Current Transfers
X. Proceeds for Invisibles Restrictions	X.A. Repatriation requirements on Proceeds
	X.A.1. Surrender Requirements on Proceeds
	X.B. Restrictions on use of funds

Source: Estefania-Flores et al. 2024. Downloaded from here.

Dataset, `.do` file, `.R` script

The replication dataset and code files are available at the following link:
`https://drive.google.com/drive/folders/13MHhNM_WL1-zoekux-dHNwU6Dajrwj8m?usp=sharing`